23.1 Mass Defect & Nuclear Binding Energy

Question Paper

Course	CIE A Level Physics
Section	23. Nuclear Physics
Topic	23.1 Mass Defect & Nuclear Binding Energy
Difficulty	Medium

Time allowed: 50

Score: /41

Percentage: /100

Question 1a

Data for a nucleus and some particles are given in Table 9.1.

Table 9.1

nucleus or particle	mass / u
$^{139}_{57}\mathrm{La}$	138.955
$^{1}_{0}\mathrm{n}$	1.00863
$^{1}_{1}\mathrm{p}$	1.00728
$^{0}_{-1}e$	5.49×10^{-4}

One nuclear reaction that can take place in a nuclear reactor may be represented, in part, by the equation shown below.

Complete the equation.

$$^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \longrightarrow {}^{95}_{42}\mathrm{Mo} + {}^{139}_{57}\mathrm{La} + 2\,{}^{1}_{0}\mathrm{n} + \ldots\ldots\ldots\ldots\ldots + \text{energy}$$

[1 mark]

Question 1b

(i)
Show that the energy equivalent to 1.00 u is 934 MeV.

[3]

(ii)
Calculate the binding energy per nucleon, in MeV, of lanthanum-139 ($^{139}_{57}\mathrm{La}$).

binding energy per nucleon = ... MeV [3]

[6 marks]

Question 1c

State and explain whether the binding energy per nucleon of uranium-235 ($^{235}_{92}\mathrm{U}$) is greater, equal to or less than the binding energy per nucleon of lanthanum-139 ($^{139}_{57}\mathrm{La}$).

[2 marks]

Question 2a

(i)

State what is meant by nuclear fission.

[1]

(ii)

On Fig. 1.1 below, sketch a line to show the variation with nucleon number *A* of the binding energy per nucleon *E* of a nucleus.

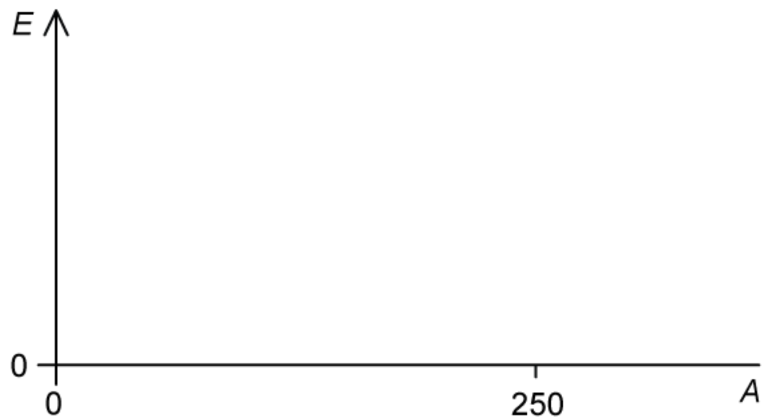


Fig. 1.1

[2]

(iii)

Explain, with reference to Fig. 1.1, why nuclear fission reactions result in the release of energy.

[2]

[5 marks]

Question 2b

A nuclear fission reaction occurs that has the following equation

$$^{1}_{0}\text{n} + ^{235}_{92}\text{U} \rightarrow ^{90}_{38}\text{Sr} + ^{143}_{54}\text{Xe} + x^{1}_{0}\text{n}$$

(i)

Determine, for this nuclear reaction, the value of x.

[1]

(ii)

Data for the binding energy per nucleon of some nuclei are given in Table 1.2.

Use the data to calculate the energy, in MeV, released in this reaction.

Table 1.2

	binding energy per nucleon / MeV
$^{235}_{92}\text{U}$	7.59
$^{90}_{38}\text{Sr}$	8.70
$^{143}_{54}\text{Xe}$	8.20

[2]

[3 marks]

Question 2c

Under the right conditions, a hydrogen-2 ^{2}H nucleus can fuse with a hydrogen-1 ^{1}H to make a helium-3 ^{3}He nucleus. The values of binding energy per nucleon for these nuclei are shown in Table 1.3.

Table 1.3

Nuclei	binding energy per nucleon / MeV
^{2}H	0.864
^{3}He	2.235

(i)

Write down the nuclear equation for this reaction.

[1]

(ii)

Explain why the binding energy per nucleon of hydrogen-1 is zero.

[1]

(iii)

Using the data in Table 1.3, calculate the energy released in this reaction.

[2]

[4 marks]

Question 2d

When two hydrogen-2 nuclei fuse into a helium-4 nucleus, 3.6×10^{-12} J of energy is released. This reaction is shown below

$$^{2}H + {}^{2}H \rightarrow {}^{4}He$$

Show that the fusion of 1 kg of two hydrogen-2 nuclei releases about 8 times more energy per kg than the fission of 1 kg of uranium-235.

The masses of hydrogen-2 and uranium-235 are shown in Table 1.4.

Table 1.4

	mass / u
$^{2}_{1}H$	2.013553
$^{235}_{92}U$	235.0439

[4 marks]

Question 3a

Define the terms

(i)
mass defect

[1]

(ii)
binding energy of a nucleus.

[2]

[3 marks]

Question 3b

If deuterium $\left({}^{2}_{1}\text{H}\right)$ nuclei undergo fusion, a possible reaction is

$$ {}^{2}_{1}\text{H} + {}^{2}_{1}\text{H} \rightarrow {}^{3}_{1}\text{H} + {}^{1}_{1}\text{H} $$

The masses of the nuclei involved in the reaction are given in Table 1.1.

Table 1.1

	mass / u
${}^{1}_{1}\text{H}$	1.0078
${}^{2}_{1}\text{H}$	2.0135
${}^{3}_{1}\text{H}$	3.0160

(i)

Explain what is meant by nuclear fusion.

[1]

(ii)

Determine the energy released, in J, when 3.00 mol of deuterium undergoes this fusion reaction.

[5]

[6 marks]

Question 3c

Fig. 1.2 shows the variation of nucleon number with the binding energy per nucleon of a nucleus.

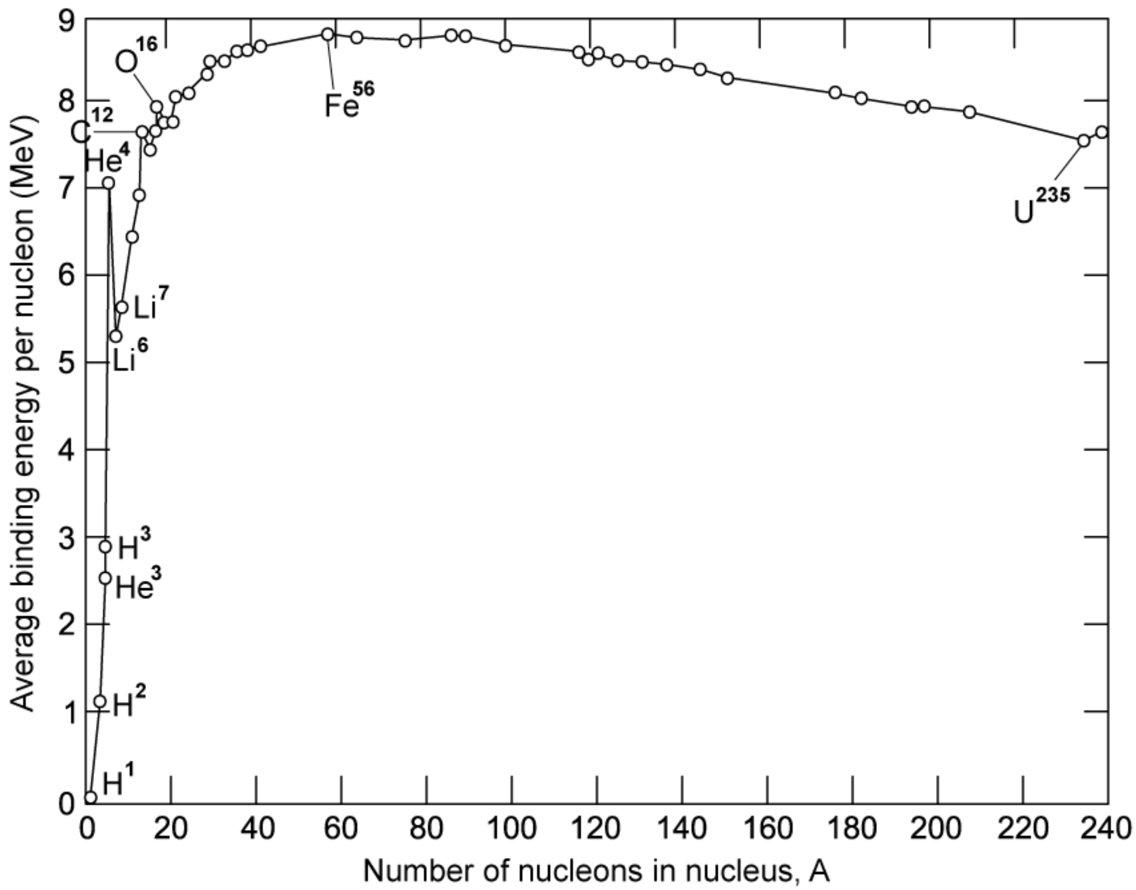


Fig. 1.2

With reference to Fig. 1.2, state and explain

(i)
which of the elements shown is the most stable

[2]

(ii)
how the graph can be used to predict whether a nucleus will undergo fusion or fission.

[2]

[4 marks]

Question 3d

Fission and fusion reactions release different amounts of energy.

Explain why the energy released per nucleon from fusion is greater than that from fission. State the feature of the graph in Fig. 1.2 that shows this.

[3 marks]

Model Answers: Medium

1a

(a) Complete the equation $^{235}_{92}U + ^{1}_{0}n \longrightarrow ^{95}_{42}Mo + ^{139}_{57}La + 2^{1}_{0}n +$ .... + energy:

Balance the nucleon numbers in the equation:

- 235 + 1 = 95 + 139 + 2 + A
- A = 236 – 95 – 139 – 2 = 0

Balance the proton numbers in the equation:

- 92 + 0 = 42 + 57 + 0 + Z
- Z = 92 – 42 – 57 = –7

The missing term in the equation in nuclide notation is:

- $7\ ^{0}_{-1}e$ [1 mark]

[Total: 1 mark]

1b

(b)

(i) Show that the energy equivalent to 1.00 u is 934 MeV:

List the known quantities:

- Mass, m = 1.00 u = 1.6 × 10^{-27} kg
- Speed of light in free space, c = 3.00 × 10^{8} m s^{-1}

Use the mass-energy equivalence equation to calculate the energy, E in J:

- $E = mc^2$ [1 mark]
- $E = (1.6 \times 10^{-27}) \times (3 \times 10^{8})^2 = 1.494 \times 10^{-10}$ J [1 mark]

Convert the energy from J to MeV:

- $E = (9.34 \times 10^{8}) \div (1.6 \times 10^{-13}) = 934$ MeV [1 mark]

(ii) Calculate the binding energy per nucleon, in MeV, of lanthanum-139 $\left({}^{139}_{57}\mathrm{La}\right)$:

List the known quantities:

- Nucleon number = 139
- Number of protons = 57
- Number of neutrons = 139 – 57 = 82
- Rest mass of a neutron, m_n = 1.00863u
- Rest mass of a proton, m_p = 1.00728u
- Mass of ${}^{139}_{57}\mathrm{La}$ = 138.955u

Calculate the mass defect, Δm of ${}^{139}_{57}\mathrm{La}$:

- $\Delta m = 82m_n + 57m_p - mass\ of\ nuclide$
- $\Delta m = (82 \times 1.00863)u + (57 \times 1.00728)u - 138.955u$
- $\Delta m = -1.16762u$ [1 mark]

Calculate the binding energy per nucleon:

- Binding energy = 1.16762 × 934 MeV = 1090 MeV [1 mark]
- Binding energy per nucleon = $\frac{binding\ energy}{nucleon\ number}$
- Binding energy per nucleon = $\frac{1.16762 \times 934}{139} = 7.84\ \mathrm{MeV}$ [1 mark]

[Total: 6 marks]

- **Memorise** the conversions between:
  - J and eV
  - u and kg
- **Read** the **question carefully** to apply the correct conversions

1c

(c) State and explain whether the binding energy per nucleon of uranium-235 ($^{235}_{92}U$) is greater, equal to or less than the binding energy per nucleon of lanthanum-139 ($^{139}_{57}La$):

State:

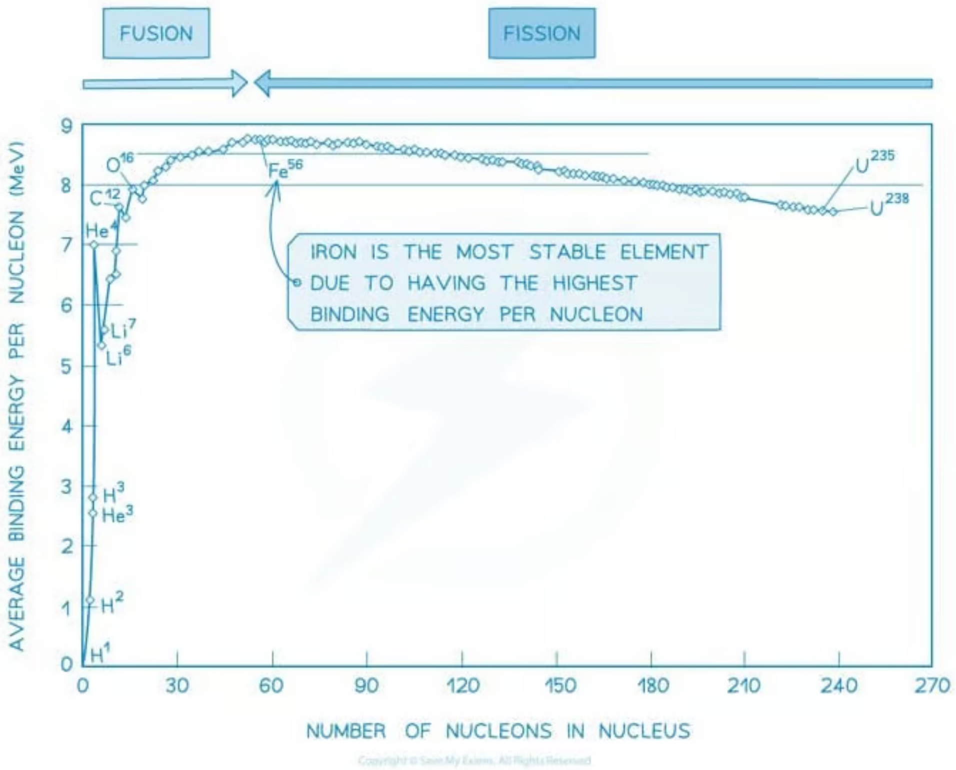
- The binding energy per nucleon of $^{235}_{92}U$ is less; [1 mark]

Explanation:

- Uranium-235 has a larger nucleon number than lanthanum-139
- So, the binding energy is split between more nucleons; [1 mark]

[Total: 2 marks]

- Remember the **binding energy per nucleon graph**, **Uranium** is the **most unstable nuclei**, so its binding energy per nucleon is the lowest for elements that undergo fission:



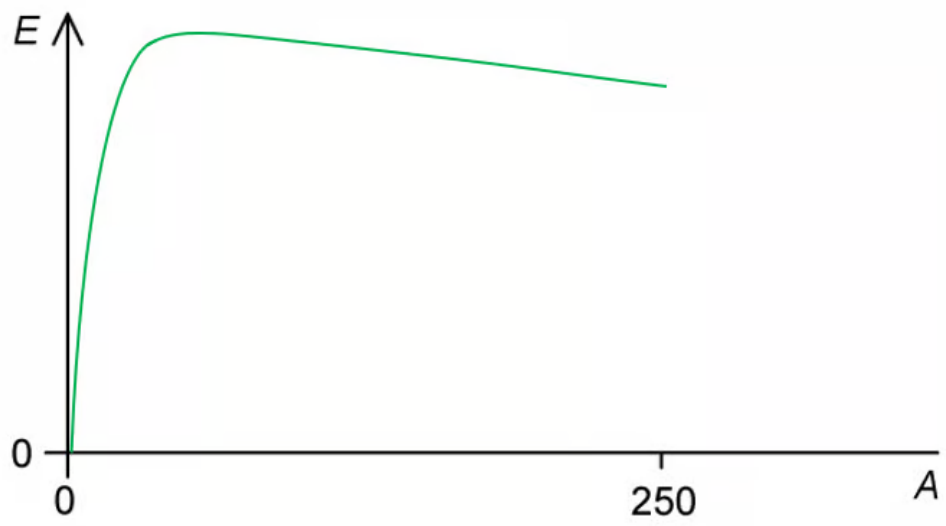
2a

(a)

(i) Nuclear fission is defined as:

- When a (single) large nucleus divides to form (smaller) nuclei; [1 mark]

(ii) The variation of nucleon number with the binding energy per nucleon is as follows:



- Curve starts at / close to the origin and forms a single peak; [1 mark]
- Peak drawn to left of centre, with a steep line on LHS and shallow line on RHS; [1 mark]

(iii) Nuclear fission reactions result in the release of energy because:

Any **two** from:

- The products of fission have a higher binding energy (per nucleon) than reactants; [1 mark]
- Large nuclei which have lower binding energy per nucleon release energy when broken apart; [1 mark]
- Nuclei (that undergo fission) have high *A* values **OR** are found on the very right side of the curve; [1 mark]

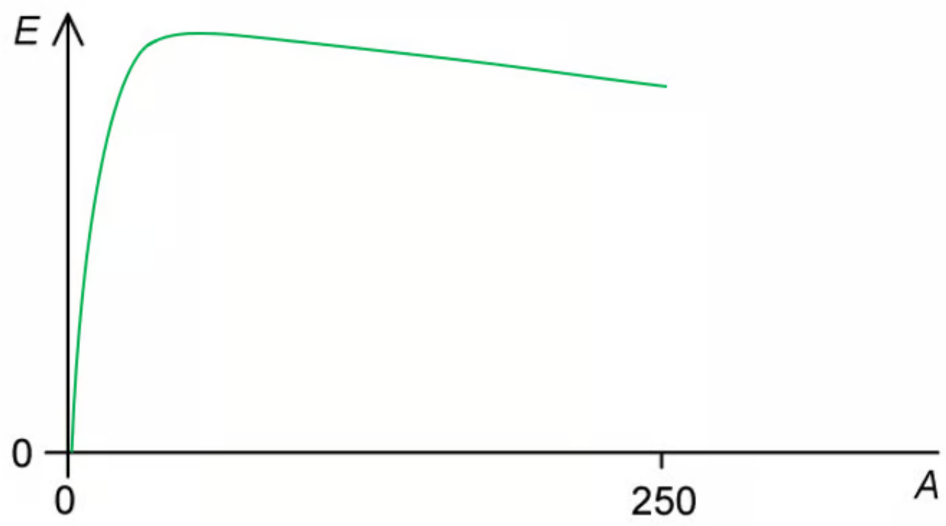
2a

(a)

(i) Nuclear fission is defined as:

- When a (single) large nucleus divides to form (smaller) nuclei; [1 mark]

(ii) The variation of nucleon number with the binding energy per nucleon is as follows:



- Curve starts at / close to the origin and forms a single peak; [1 mark]
- Peak drawn to left of centre, with a steep line on LHS and shallow line on RHS; [1 mark]

(iii) Nuclear fission reactions result in the release of energy because:

Any **two** from:

- The products of fission have a higher binding energy (per nucleon) than reactants; [1 mark]
- Large nuclei which have lower binding energy per nucleon release energy when broken apart; [1 mark]
- Nuclei (that undergo fission) have high *A* values **OR** are found on the very right side of the curve; [1 mark]

- Large nuclei which have lower binding energy per nucleon release energy when broken apart; [1 mark]
- Nuclei (that undergo fission) have high *A* values **OR** are found on the very right side of the curve; [1 mark]

[Total: 5 marks]

2b

(b)

(i) The value of x is:

- ${}^{1}_{0}\text{n} + {}^{235}_{92}\text{U} \rightarrow {}^{90}_{38}\text{Sr} + {}^{143}_{54}\text{Xe} + x\,{}^{1}_{0}\text{n}$
- Number of nucleons on LHS: $1 + 235 = 236$
- Number of nucleons on RHS: $90 + 143 + x = 233 + x$
- Hence, $x = 236 - 233 = 3$ [1 mark]

(ii) Calculate the energy change that takes place in this reaction.

List the known quantities:

- Binding energy per nucleon of U-235, $E_U = 7.59$ MeV
- Binding energy per nucleon of Sr-90, $E_S = 8.70$ MeV
- Binding energy per nucleon of Xe-143, $E_X = 8.20$ MeV

The amount of energy released during the fission reaction:

- Energy released: $\Delta E = (90E_S + 143E_X) - 235E_U$
- Energy released: $\Delta E = (90 \times 8.70) + (143 \times 8.20) - (235 \times 7.59)$ [1 mark]
- Energy released: $\Delta E = 172$ MeV [1 mark]

[Total: 3 marks]

2c

(c)

(i) The nuclear equation for this reaction is:

- $^{2}H + ^{1}H \rightarrow ^{3}He$ [1 mark]

(ii) The binding energy per nucleon of hydrogen-1 is zero because:

- Hydrogen-1 is a single proton, so it is not bound to any other nucleons / it cannot release energy (by separation); [1 mark]

(iii) Calculate the energy released in this reaction:

List the known quantities:

- Binding energy per nucleon of hydrogen-2, E_H = 0.864 MeV
- Binding energy per nucleon of helium-3, E_{He} = 2.235 MeV

The amount of energy released during the fission reaction:

- Energy released: $\Delta E = 3E_{He} - 2E_H$
- Energy released: ΔE = (3 × 2.235) – (2 × 0.864) [1 mark]
- Energy released: ΔE = 5.0 MeV [1 mark]

[Total: 4 marks]

2d

(d) Determine the energy per kg for the fission and fusion reactions:

List the known quantities:

- Mass of ^{2}H = 2.013553 u

- Mass of ^{235}U = 235.0439 u
- Atomic mass unit, $u = 1.66 \times 10^{-27}$ kg
- Energy available from fusion of ^{2}H = 3.6×10^{-12} J
- Energy available from fission of ^{235}U = 172 MeV (from **(b)**)

Calculate the energy available per 1 kg of hydrogen-2:

- Mass of <u>two</u> ^{2}H nuclei = 2 × 2.013553 u
- Number of ^{2}H nuclei in 1 kg = $\dfrac{1 \text{ kg}}{2 \times 2.013553 \times (1.66 \times 10^{-27})}$ = 1.496×10^{26} [1 mark]
- Energy per kg = $(1.496 \times 10^{26}) \times (3.6 \times 10^{-12}) = 5.39 \times 10^{14}$ J [1 mark]

Calculate the energy available per 1 kg of uranium-235:

- Mass of <u>one</u> ^{235}U nucleus = 235.0439 u
- Number of ^{235}U nuclei in 1 kg = $\dfrac{1 \text{ kg}}{235.0439 \times (1.66 \times 10^{-27})}$ = 2.562×10^{24}
- Convert energy to J: 172 MeV = $(172 \times 10^{6}) \times (1.6 \times 10^{-19})$
- Energy per kg = $(2.562 \times 10^{24}) \times (172 \times 10^{6}) \times (1.6 \times 10^{-19}) = 7.05 \times 10^{13}$ J [1 mark]

Compare the two values for energy released per kg:

- Ratio: $\dfrac{\textit{energy per kg from fusion}}{\textit{energy per kg from fission}} = \dfrac{5.39 \times 10^{14}}{7.05 \times 10^{13}}$ = 7.65 ≈ 8 [1 mark]
- Hence, hydrogen fusion provides about 8 times more energy per kg than uranium fission

[Total: 4 marks]

3a

(a)

(i) Mass defect is defined as:

- The difference between the mass of the nucleus and the sum of the masses of its nucleons if separated completely; [1 mark]

(ii) Binding energy of a nucleus is defined as:

Any **pair** from:

- The (minimum) energy required / work done to separate the nucleons (in a nucleus); [1 mark]
- To infinity; [1 mark]
  OR
- The energy released when nucleons come together (to form a nucleus); [1 mark]
- From infinity; [1 mark]

[Total: 3 marks]

3b

(b)

(i) Fusion is defined as:

- When two nuclei combine to form a (single) nucleus; [1 mark]

(ii) Determine the energy released when 3.00 mol of deuterium reacts:

List the known quantities:

- Mass of hydrogen-1, m_1 = 1.0078 u
- Mass of hydrogen-2, m_2 = 2.0135 u
- Mass of hydrogen-3, m_3 = 3.0160 u
- Atomic mass unit, $u = 1.66 \times 10^{-27}$ kg
- Speed of light, $c = 3.00 \times 10^{8}$ m s^{-1}
- Avogadro constant, $N_A = 6.02 \times 10^{23}$ mol^{-1}

Determine the mass defect:

- Mass defect: $\Delta m = 2m_2 - (m_3 + m_1)$
- Δm = [(2 × 2.0135) – (3.0160 + 1.0078)] u
- Δm = 0.0032 u [1 mark]

- The energy released when nucleons come together (to form a nucleus); [1 mark]
- From infinity; [1 mark]

[Total: 3 marks]

3b

(b)

(i) Fusion is defined as:

- When two nuclei combine to form a (single) nucleus; [1 mark]

(ii) Determine the energy released when 3.00 mol of deuterium reacts:

List the known quantities:

- Mass of hydrogen-1, m_1 = 1.0078 u
- Mass of hydrogen-2, m_2 = 2.0135 u
- Mass of hydrogen-3, m_3 = 3.0160 u
- Atomic mass unit, $u = 1.66 \times 10^{-27}$ kg
- Speed of light, $c = 3.00 \times 10^8$ m s^{-1}
- Avogadro constant, $N_A = 6.02 \times 10^{23}$ mol^{-1}

Determine the mass defect:

- Mass defect: $\Delta m = 2m_2 - (m_3 + m_1)$
- $\Delta m = [(2 \times 2.0135) - (3.0160 + 1.0078)]\ u$
- $\Delta m = 0.0032$ u [1 mark]

Use mass-energy equivalence to determine the energy released:

- Mass-energy equivalence: $\Delta E = \Delta mc^2$ [1 mark]
- $\Delta E = 0.0032 \times (1.66 \times 10^{-27}) \times (3.00 \times 10^8)^2$

- ΔE = 4.78 × 10^{-13} J [1 mark]

Determine the energy released when 3.00 mol of deuterium reacts:

- 3.00 mol of deuterium forms 1.50 mol of hydrogen-3 [1 mark]
- Therefore, the number of hydrogen-3 nuclei = 1.50 mol × N_A
- Total energy = 1.50 × (6.02 × 10^{23}) × (4.78 × 10^{-13})
- Total energy = 4.32 × 10^{11} J [1 mark]

[Total: 6 marks]

3c

(c)

(i) The most stable element is:

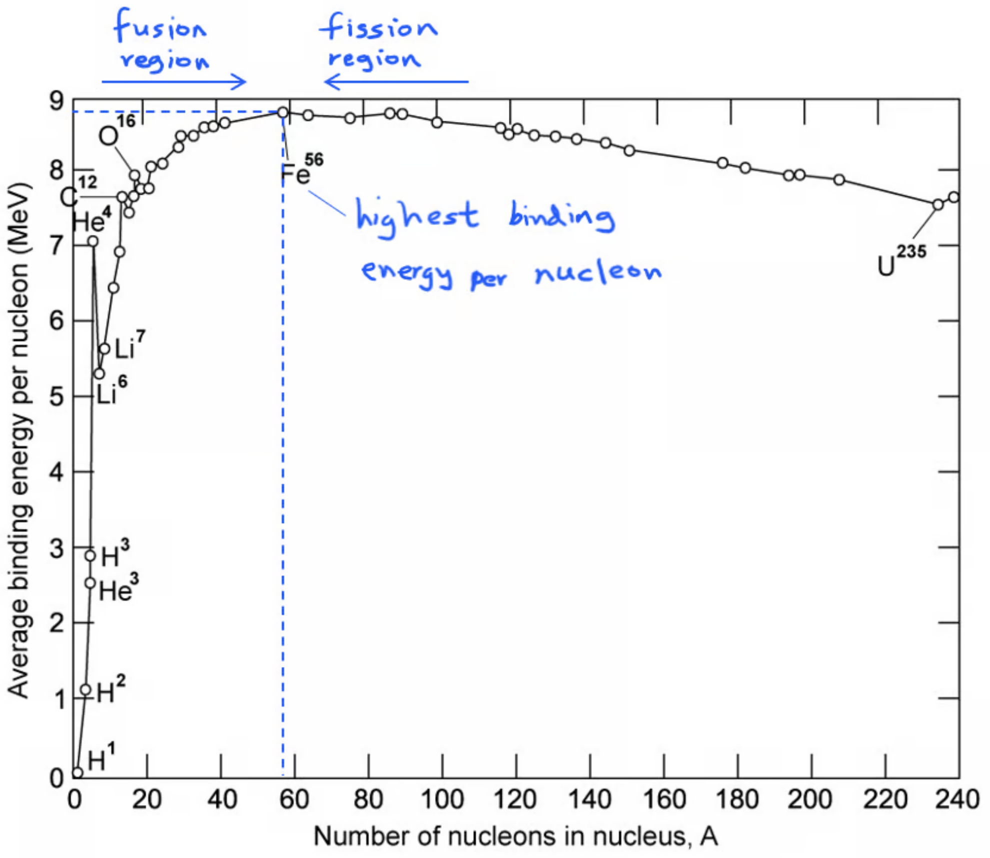
- Iron / Fe-56; [1 mark]
- (Because) it has the highest binding energy per nucleon; [1 mark]

(ii) The graph can be used to predict whether a nucleus will undergo fusion or fission because:

- Fusion occurs at *A* values (nucleon number) below the peak / lighter than Fe-56; [1 mark]
- Fission occurs at *A* values above the peak / heavier than Fe-56; [1 mark]

[Total: 4 marks]

You could also score the marks by clearly annotating the graph



Remember:

- Fusion is the joining of two nuclei (into one larger one)
- Hence the reactants will have a higher binding energy per nucleon than the resulting nuclei
- This can only happen at small values of *A*

- Fission is the splitting of a nucleus (into two smaller ones)
- Hence the reactant will have a lower binding energy per nucleon than its daughter nuclei
- This can only happen at large values of *A*

3d

(d) More energy is released per nucleon from fusion because:

- Energy is released when a nucleus forms with a higher binding energy (per nucleon); [1 mark]

- (When energy is released from fusion) the increase in binding energy (per nucleon) is greater than the increase in binding energy (per nucleon) from fission;

[1 mark]

- (The graph shows this) at small values of A (fusion region), the gradient is much steeper compared to the gradient at large values of A (fission region); [1 mark]